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#include <stdio.h>
#include <stdlib.h>
struct Node{
    int data;
    struct Node* next;
};
struct Node* top=NULL;

void push(int val)
{
    struct Node* newNode=(struct Node*)malloc(sizeof(struct Node));

    if(newNode==NULL)
    {
        printf("stack overflow");
    }
    newNode->data=val;
    newNode->next=top;
    top=newNode;
}

void pop()
{
    if(top==NULL)
    {
        printf("stack underflow");
        return;
    }
    struct Node* temp=top;
    top=top->next;
    free(temp);
}

void display()
{
    if(top==NULL)
    {
        printf("\nStack Empty ");
        return;
    }
    struct Node* temp=top;
    printf("Stack [TOP]: ");
    while(temp!=NULL)
    {
        printf("%d ]",temp->data);
        temp=temp->next;
    }
    printf("\n");
}

```

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}

int main()
{
    int choice,x;
    while(1){
        printf("\n 1.Push \n 2.Pop \n 3.Display \n 4.Exit");
        printf("\n enter choice:");
        scanf("%d",&choice);
        switch(choice){

            case 1:
                printf("\n enter the element to be inserted:");
                scanf("%d",&x);
                push(x);
                break;

            case 2:
                pop();

                break;

            case 3:
                display();
                break;

            case 4:
                return 0;
                break;

            default:
                printf("\ninvalid choice");
        }
    }
    return 0;
}

```